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Skill Session 4 & 5

Step 1:Created the database and inserted the data

Organization: Tejaswini's Org - 202... Project: Project 0

Database: bankdb Collection: customers

Documents: 8 Aggregations: 0 Schema: 1 Indexes: 1 Validation: 0

Query: Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

Type a query: { field: 'value' }

Buttons: ADD DATA, BULK, EXPORT CODE

customers	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status String
1	ObjectId('6a7e8b8dbc6847...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2	ObjectId('6a7e8b8dbc6847...')	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
3	ObjectId('6a7e8c29bc6847...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
4	ObjectId('6a7e8c29bc6847...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
5	ObjectId('6a7e8c3bbc6847...')	105	"Kiran Kumar"	"Guntur"	"Current"	65000	"Inactive"
6	ObjectId('6a7e8c56bc6847...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
7	ObjectId('6a7e8c66bc6847...')	107	"Rahul Verma"	"Bengaluru"	"Current"	95000	"Active"
8	ObjectId('6a7e8c72bc6847...')	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

Organization: Tejaswini's Org - 202... Project: Project 0

Database: bankdb Collection: transactions

Documents: 15 Aggregations: 0 Schema: 1 Indexes: 1 Validation: 0

Query: Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

Type a query: { field: 'value' }

Buttons: ADD DATA, BULK, EXPORT CODE

transactions	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode String	date String
1	ObjectId('6a7efed2bc6847...')	1001	101	"Deposit"	25000	"UPI"	"2026-08-01"
2	ObjectId('6a7efee0bc6847...')	1002	101	"Withdrawal"	10000	"ATM"	"2026-08-02"
3	ObjectId('6a7efefebc6847...')	1003	101	"Withdrawal"	10000	"ATM"	"2026-08-02"
4	ObjectId('6a7ef8f10bc6847...')	1004	102	"Withdrawal"	15000	"ATM"	"2026-08-03"
5	ObjectId('6a7ef8f22bc6847...')	1005	103	"Deposit"	30000	"UPI"	"2026-08-03"
6	ObjectId('6a7ef8f30bc6847...')	1006	103	"Withdrawal"	5000	"ATM"	"2026-08-04"
7	ObjectId('6a7ef8f3abc6847...')	1007	104	"Deposit"	100000	"NEFT"	"2026-08-04"
8	ObjectId('6a7ef8f4abc6847...')	1008	104	"Withdrawal"	25000	"ATM"	"2026-08-05"
9	ObjectId('6a7ef8f56bc6847...')	1009	105	"Deposit"	15000	"UPI"	"2026-08-05"
10	ObjectId('6a7ef8f63bc6847...')	1010	106	"Deposit"	75000	"NEFT"	"2026-08-06"
11	ObjectId('6a7ef8f6fbc6847...')	1011	106	"Withdrawal"	20000	"ATM"	"2026-08-06"
12	ObjectId('6a7ef8f7ebc6847...')	1012	107	"Deposit"	40000	"IMPS"	"2026-08-07"
13	ObjectId('6a7ef8f8bc6847...')	1013	107	"Withdrawal"	12000	"ATM"	"2026-08-07"
14	ObjectId('6a7ef8f98bc6847...')	1014	108	"Deposit"	150000	"NEFT"	"2026-08-08"
15	ObjectId('6a7ef8fa7bc6847...')	1015	108	"Withdrawal"	30000	"ATM"	"2026-08-08"

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Step 2: Finding the Active Status

The screenshot shows the Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-items 'admin', 'bankdb', 'customers', 'transactions', 'bookflow', 'local', and 'sample_mflix'. The main area displays the 'customers' cluster. A query editor at the top shows a JSON query: `{ "status": "Active" }`. Below the query editor, a table of customer data is displayed. The table has columns: `_id`, `ObjectID`, `customer_id`, `name`, `city`, `account_type`, `balance`, and `status`. The data is filtered to show only active customers.

<code>_id</code>	<code>ObjectID</code>	<code>customer_id</code>	<code>name</code>	<code>city</code>	<code>account_type</code>	<code>balance</code>	<code>status</code>
1	<code>ObjectID('6a7e8b8db6c6847...')</code>	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2	<code>ObjectID('6a7e8b8db6c6847...')</code>	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
3	<code>ObjectID('6a7e8b8db6c6847...')</code>	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
4	<code>ObjectID('6a7e8c29bc6847...')</code>	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
5	<code>ObjectID('6a7e8c56bc6847...')</code>	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
6	<code>ObjectID('6a7e8c56bc6847...')</code>	107	"Rahul Verma"	"Bengaluru"	"Current"	95000	"Active"
7	<code>ObjectID('6a7e8c73bc6847...')</code>	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

Step 3: To find balance greater than 100000

The screenshot shows the Data Explorer interface with a new query: `{ "balance": { "$gt": 100000 } }`. The table below shows the results of this query, filtered to show only customers with a balance greater than 100,000.

<code>_id</code>	<code>ObjectID</code>	<code>customer_id</code>	<code>name</code>	<code>city</code>	<code>account_type</code>	<code>balance</code>	<code>status</code>
1	<code>ObjectID('6a7e8b8db6c6847...')</code>	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
2	<code>ObjectID('6a7e8c29bc6847...')</code>	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
3	<code>ObjectID('6a7e8c56bc6847...')</code>	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
4	<code>ObjectID('6a7e8c73bc6847...')</code>	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

Step 4: To find Hyderabad Customers

ORGANIZATION
Tejaswini's Org - 202...

PROJECT
Project 0

Compare tiers

book_metadata

transactions

customers

bankdb

+

book_metadata

transactions

customers

bankdb

+

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

{

"city": "Hyderabad"

}

Generate

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 3 of 3

customers

_id	ObjectId	customer_id	name	city	account_type	balance	status
1	ObjectId('6a7e8b8dc6847...')	181	Ravi Kumar	Hyderabad	Savings	85000	Active
2	ObjectId('6a7e8c83bc6847...')	183	Arjun Reddy	Hyderabad	Savings	45000	Active
3	ObjectId('6a7e8c56bc6847...')	186	Anjali Singh	Hyderabad	Savings	175000	Active

Step 5: To find Savings Account Customers

ORGANIZATION
Tejaswini's Org - 202...

PROJECT
Project 0

Compare tiers

book_metadata

transactions

customers

bankdb

+

book_metadata

transactions

customers

bankdb

+

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

{

"account_type": "Savings"

}

Generate

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

customers

_id	ObjectId	customer_id	name	city	account_type	balance	status
1	ObjectId('6a7e8b8dc6847...')	181	Ravi Kumar	Hyderabad	Savings	85000	Active
2	ObjectId('6a7e8c83bc6847...')	183	Arjun Reddy	Hyderabad	Savings	45000	Active
3	ObjectId('6a7e8c29bc6847...')	184	Sneha Rao	Marangal	Savings	210000	Active
4	ObjectId('6a7e8c56bc6847...')	186	Anjali Singh	Hyderabad	Savings	175000	Active
5	ObjectId('6a7e8c72bc6847...')	188	Meena Devi	Chennai	Savings	320000	Active

Step 6: To find Transactions Greater than 20000

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar displays the cluster hierarchy: Cluster0 > banklab > transactions. The main panel shows the 'Documents' tab for the 'transactions' collection. A query is entered in the search bar: `{ "amount": { "$gt": 20000 } }`. Below the query, a table displays the results of the query, showing 8 transactions with amounts greater than 20000. The table columns are: _id, transaction_id, customer_id, type, amount, mode, and date. The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales'.

	_id	transaction_id	customer_id	type	amount	mode	date
1	ObjectId('6a7e8ed2bc6847...')	1001	181	"Deposit"	25000	"UPI"	"2026-08-01"
2	ObjectId('6a7e8f22bc6847...')	1005	183	"Deposit"	38000	"UPI"	"2026-08-03"
3	ObjectId('6a7e8f3ebc6847...')	1007	184	"Deposit"	180000	"NEFT"	"2026-08-04"
4	ObjectId('6a7e8f4abc6847...')	1008	184	"Withdrawal"	25000	"ATM"	"2026-08-05"
5	ObjectId('6a7e8f63bc6847...')	1010	185	"Deposit"	75000	"NEFT"	"2026-08-06"
6	ObjectId('6a7e8f7ebc6847...')	1012	187	"Deposit"	48000	"IMPS"	"2026-08-07"
7	ObjectId('6a7e8f99bc6847...')	1014	188	"Deposit"	150000	"NEFT"	"2026-08-08"
8	ObjectId('6a7e8fa7bc6847...')	1015	188	"Withdrawal"	38000	"ATM"	"2026-08-08"

Step 7: To find Deposits

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar displays the cluster hierarchy: Cluster0 > banklab > transactions. The main panel shows the 'Documents' tab for the 'transactions' collection. A query is entered in the search bar: `{ "type": "Deposit" }`. Below the query, a table displays the results of the query, showing 7 transactions that are deposits. The table columns are: _id, transaction_id, customer_id, type, amount, mode, and date. The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales'.

	_id	transaction_id	customer_id	type	amount	mode	date
1	ObjectId('6a7e8ed2bc6847...')	1001	181	"Deposit"	25000	"UPI"	"2026-08-01"
2	ObjectId('6a7e8f22bc6847...')	1005	183	"Deposit"	38000	"UPI"	"2026-08-03"
3	ObjectId('6a7e8f3ebc6847...')	1007	184	"Deposit"	180000	"NEFT"	"2026-08-04"
4	ObjectId('6a7e8f5abc6847...')	1009	185	"Deposit"	15000	"UPI"	"2026-08-05"
5	ObjectId('6a7e8f63bc6847...')	1010	185	"Deposit"	75000	"NEFT"	"2026-08-06"
6	ObjectId('6a7e8f7ebc6847...')	1012	187	"Deposit"	48000	"IMPS"	"2026-08-07"
7	ObjectId('6a7e8f99bc6847...')	1014	188	"Deposit"	150000	"NEFT"	"2026-08-08"

Step 8: To find ATM transactions

The screenshot shows the Data Explorer interface with the following components:

- Left Panel:** Contains 'Data Explorer', 'My Queries', 'Data Modeling', and 'CLUSTERS [0]'. The 'CLUSTERS' section shows a tree view with 'Cluster0' containing 'admin', 'bankdb', 'customers', and 'transactions'.
- Top Bar:** Displays 'ORGANIZATION: Tejswini's Org - 202...', 'PROJECT: Project 0', and a 'Compare tiers' button.
- Breadcrumbs:** 'Cluster0 > bankdb > transactions'.
- Query Editor:** A text box containing the JSON query:

```
{ "mode": "ATM" }
```

. Buttons for 'Explain', 'Reset', 'Find', and 'Options' are present.
- Table:** A table with 8 rows of transaction data. The columns are: `_id` (ObjectID), `transaction_id` (Int32), `customer_id` (Int32), `type` (String), `amount` (Int32), `mode` (String), and `date` (String). The data shows various withdrawal transactions.
- Bottom Bar:** Includes 'ADD DATA', 'BULK', and 'EXPORT CODE' buttons, along with pagination '26' and '1 - 8 of 8'.

Step 9: To find Deposits greater than 50000

The screenshot shows the Data Explorer interface with the following components:

- Left Panel:** Similar to Step 8, showing the 'Data Explorer' and 'CLUSTERS' sidebar.
- Top Bar:** Same as Step 8, showing organization and project information.
- Breadcrumbs:** 'Cluster0 > bankdb > transactions'.
- Query Editor:** A text box containing the JSON query:

```
{ "type": "Deposit", "amount": { "$gt": 50000 } }
```

. Buttons for 'Explain', 'Reset', 'Find', and 'Options' are present.
- Table:** A table with 3 rows of transaction data. The columns are: `_id` (ObjectID), `transaction_id` (Int32), `customer_id` (Int32), `type` (String), `amount` (Int32), `mode` (String), and `date` (String). The data shows three deposit transactions with amounts 180000, 75000, and 150000.
- Bottom Bar:** Includes 'ADD DATA', 'BULK', and 'EXPORT CODE' buttons, along with pagination '26' and '1 - 3 of 3'.

Step 10: To find Hyderabad Customers with high balance

The screenshot shows the Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bankdb', 'customers', 'transactions', 'bookflow', 'local', and 'sample_mflix'. The main panel is titled 'Cluster0 > bankdb > customers'. It shows a query editor with a JSON query:

```
{ "city": "Hyderabad", "balance": { "$gt": 100000 } }
```

. Below the query editor, a table displays the results. The table has columns: _id, customer_id, name, city, account_type, balance, and status. The results show one customer: Anjali Singh, Hyderabad, Savings, 175000, Active.

_id	customer_id	name	city	account_type	balance	status
ObjectID('6a7e8c56bc6847...')	106	Anjali Singh	Hyderabad	Savings	175000	Active

Step 11: To find Customers in Multiple Cities

The screenshot shows the Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bankdb', 'customers', 'transactions', 'bookflow', 'local', and 'sample_mflix'. The main panel is titled 'Cluster0 > bankdb > customers'. It shows a query editor with a JSON query:

```
{ "city": { "$in": [ "Hyderabad", "Vijayawada", "Warangal" ] } }
```

. Below the query editor, a table displays the results. The table has columns: _id, customer_id, name, city, account_type, balance, and status. The results show five customers: Ravi Kumar (Hyderabad, Savings, 85000, Active), Priya Sharma (Vijayawada, Current, 125000, Active), Arjun Reddy (Hyderabad, Savings, 45000, Active), Sneha Rao (Warangal, Savings, 210000, Active), and Anjali Singh (Hyderabad, Savings, 175000, Active).

_id	customer_id	name	city	account_type	balance	status
ObjectID('6a7e8b8dbcc6847...')	101	Ravi Kumar	Hyderabad	Savings	85000	Active
ObjectID('6a7e8b8dbcc6847...')	102	Priya Sharma	Vijayawada	Current	125000	Active
ObjectID('6a7e8c83bc6847...')	103	Arjun Reddy	Hyderabad	Savings	45000	Active
ObjectID('6a7e8c29bc6847...')	104	Sneha Rao	Warangal	Savings	210000	Active
ObjectID('6a7e8c56bc6847...')	106	Anjali Singh	Hyderabad	Savings	175000	Active

Step 12: Sort Customers by Balance

ORGANIZATION: Tejswini's Org - 202... PROJECT: Project 0

Compare tiers

Data Explorer

My Queries

Data Modelling

CLUSTERS {}

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- bookflow
- local
- sample_mflix

book_metadata transactions customers bankdb

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Sort

Generate aggregation + Explain Run Options

Preview

STAGES TEXT WIZARD

Sort

balance: 85000 status: "Active" balance: 125000 status: "Active" balance: 45000 status: "Active" balance: 210000 status: "Active"

Stage 1 \$sort

```
1 * {
2   "balance": -1
3 }
4
```

Output preview after \$sort stage (Sample of 8 documents)

```
{ "_id": ObjectId("6a7e8c72bc6847edec8995d4"), "customer_id": 100, "name": "Mrunal Dev", "city": "Chennai", "account_type": "Savings", "balance": 325000, "status": "Active" }
{ "_id": ObjectId("6a7e8c29bc6847edec8995d5"), "customer_id": 104, "name": "Srinha Rao", "city": "Warangal", "account_type": "Savings", "balance": 210000, "status": "Active" }
{ "_id": ObjectId("6a7e8c56bc6847edec8995d6"), "customer_id": 106, "name": "Anjali Singh", "city": "Hyderabad", "account_type": "Savings", "balance": 175000, "status": "Active" }
```

+ Add stage

Learn more about aggregation pipeline stages

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Step 13: Calculate Total Transaction amount

ORGANIZATION: Tejswini's Org - 202... PROJECT: Project 0

Compare tiers

Data Explorer

My Queries

Data Modelling

CLUSTERS {}

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- bookflow
- local
- sample_mflix

book_metadata transactions customers bankdb

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

\$group

Generate aggregation + Explain Run Options

Preview

STAGES TEXT WIZARD

\$group

balance: 85000 status: "Active" balance: 125000 status: "Active" balance: 45000 status: "Active" balance: 210000 status: "Active"

Stage 1 \$group

```
1 {
2   "_id": null,
3   "total_amount": {
4     "$sum": "$amount"
5   }
6 }
7
```

Output preview after \$group stage (Sample of 1 document)

```
{ "_id": null, "total_amount": 0 }
```

+ Add stage

Learn more about aggregation pipeline stages

Alert System Status: All Good

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